

Healthy Skepticism · Session 4 take-home summary

The fault: the causal leap

Two things move together, and the headline says one causes the other. That is the fifth and last fault of the course, and it is the one the first four were building toward. A correlation is a fact about two columns of numbers. A cause is a claim about what would happen if you changed one of them. The gap between those is where almost every health headline lives.

The session sorted every strong correlation into one of three cases, and gave you a question to ask for each. The three questions, in order, are the whole day.

Question 1. Can you draw the grid of people?

Before anything else, ask what one square in the 100-person grid would be. If the answer is "a year," there are no people in the correlation and nothing to explain.

Two examples from Tyler Vigen's Spurious Correlations project. Saturn's distance from the sun tracks the number of physical-science degrees awarded in the United States, $r = 0.987$. American cheese consumption per person tracks BlackRock's share price, $r = 0.94$. Both would clear any journal's significance bar. Both are found by trying millions of pairs of series and keeping the ones that happen to line up, and both are trends measured in years, so there is no person who is both a cheese-eater and a share price. Take the year-to-year changes instead of the running totals and the cheese correlation falls to 0.29.

Vigen T, Spurious Correlations, tylervigen.com, CC BY 4.0. Correlations #2,656 and #4,018; the year-to-year figure is calculated from his data.

Case 1, coincidence. Search enough pairs and a near-perfect match turns up somewhere, meaning nothing. Any two things that grow will correlate.

Question 2. Could something else drive both columns?

Once there are real people, the correlation is real too, and the question becomes whether a third thing sits behind both columns. We drew that as a fork: one box at the top with an arrow into each of the two things that moved together, and a dashed line between them where the headline's arrow was. Three examples.

The sugar pill. In the Coronary Drug Project, men in the placebo arm who took their pill faithfully had 15.1% five-year mortality; those who did not had 28.3%. The pill was sugar. The gap is carried entirely by the kind of person who follows instructions, and it is as large as many real drug effects.

Coronary Drug Project Research Group. N Engl J Med 1980;303:1038-1041.

Bird watching and dementia. An illustrative headline: bird watchers show lower rates of dementia. The fork has good cognition at the top. People whose minds are still sharp go out with binoculars, and they are the same people who are not diagnosed. Whitehall II followed 10,308 civil servants for 28 years: activity made no difference to dementia risk (hazard ratio 1.00), and activity fell away in the nine or ten years before diagnosis, when cognition was already declining and took the hobby with it.

Sabia S et al. BMJ 2017;357:j2709. Hazard ratio for meeting activity guidelines 1.00, 95% CI 0.80 to 1.24.

Vaccines and autism. The CDC counted about 1 child in 150 as autistic in 2000 and 1 in 31 in 2022, and the childhood vaccine schedule grew over the same years. The fork here has two things at the top: contact with doctors, and a diagnostic category that kept widening. Children who see doctors more are both vaccinated on schedule and more likely to be evaluated. And the category itself moved: in Denmark, a 1994 change in criteria and the 1995 addition of outpatient diagnoses together explain about 60% of the rise in recorded autism (95% CI 33 to 87%); CDC sites in a single year ranged fivefold, from 9.7 to 53.1 per 1,000, under identical criteria; in British Columbia at least a third of the increase was children moved in from another special-education category.

Shaw KA et al. MMWR Surveill Summ 2025;74(2):1-22. Hansen SN, Schendel DE, Parner ET. JAMA Pediatr 2015;169(1):56-62. Coo H et al. J Autism Dev Disord 2008;38(6):1036-1046.

The studies that settled it compared vaccinated and unvaccinated children in the same registry and the same era, so there was no trend left to confound. Madsen 2002 followed 537,303 Danish children (relative risk of autism after MMR 0.92, 95% CI 0.68 to 1.24). Hviid 2019 followed 657,461 (hazard ratio 0.93, 0.85 to 1.02, high-risk siblings included). Andersson 2025 followed 1.2 million for aluminum-adjuvanted vaccines (0.93, 0.88 to 1.03 for neurodevelopmental disorders). Child against child, the diagnosis arrives at the same rate.

Madsen KM et al. N Engl J Med 2002;347:1477-1482. Hviid A et al. Ann Intern Med 2019;170:513-520. Andersson NW et al. Ann Intern Med 2025;178:1369-1377. Also Taylor LE et al. Vaccine 2014;32:3623-3629, a meta-analysis of 1.26 million children, odds ratio 0.99.

Case 2, a confounder. In each, something about the people produces the correlation. The causal link is an illusion, and counting the two groups can never see past it.

Question 3. How big would the rival have to be?

Smoking and lung cancer is the session's one genuine cause, and the point of including it is that calling something a cause is hard even when it is true. The 1950s cohorts found smokers got lung cancer far more often. R. A. Fisher, the inventor of the significance test, objected: perhaps a gene drives both the craving and the cancer. That is a fork, exactly like the ones above, and it fits the two-column data just as well as the headline arrow does. He was a paid consultant to the tobacco industry, and a smoker, but the objection was legitimate and deserved an answer.

The answer used Session 2's tools twice. First the direction: among 100 British men of the era, about 65 smoked; about 10 of those and 1 of the 35 never-smokers got lung cancer (counts rounded to whole men). The forward probability, cancer given cigarettes, is 10 in 65, about 16%, against 2% for never-smokers, eight times apart. The reverse, cigarettes given cancer, is 10 in 11, 91%, and it is that high only because most men smoked. The claim is the forward direction, and it needs the prevalence of smoking to read.

Doll R. Br J Cancer 1953;7:303 (about 65% of British men smoked cigarettes around 1950). Crispo A et al. Br J Cancer 2004;91:1280-1286 (lifetime risk by age 75 about 15.7% for continuing smokers; the 2% for never-smokers is approximate).

Then the rival. Jerome Cornfield showed in 1959 that for a gene to produce an eightfold risk on its own, it would have to be at least eight times as common among smokers as among never-smokers. That is a prevalence, and it can be checked. If one never-smoker in a hundred carried it, eight smokers in a hundred would have to (illustrative numbers; only the ratio is forced). No gene like that exists, and

studies of identical twins who differed in smoking later confirmed that the smoking twin carried the excess risk.

Cornfield J et al. J Natl Cancer Inst 1959;22:173-203. Hjelmberg J et al. Thorax 2017;72:1021-1027.

Case 3, a cause. The arrow survives because the arithmetic was done, not because the correlation was strong. The rival was ruled out by a prevalence it would need and does not have.

Why the leap is a leap

Look at what each question needed. The correlation itself comes from the two columns alone; Saturn and the cheese had it. The direction needed a prevalence, how common smoking was. Ruling out the rival needed another prevalence, how common the gene would have to be. Every step past the correlation needed a number the correlation never contained. That is the whole course in one sentence: the probability reversal of Session 2 is not a separate fault from the causal leap, it is the tool that closes it.

What to ask

- What would one square in the grid be? If it is a year, stop.
- Who are the people, and what else is true of the ones in the exposed column? Faithful pill takers, bird watchers, and fully vaccinated children are all a kind of person.
- Draw the fork. What would have to sit at the top for the headline arrow to be an illusion, and does the study design rule that out?
- Is the claim the forward probability, and what prevalence does it take to read it?
- If a rival cause has been named, how big would it have to be? Has anyone checked?

Session 4 slides and the whitepapers on spurious correlation and on autism as a diagnostic category are in the library at healthy-skepticism.com.